

Hospital Economic Implications of False Positive Blood Cultures



Publication

Attributable
Incremental Hospital
Costs

Extended
Length of Stay

Infection Control and Hospital Epidemiology
(2022)¹

\$2,853*

1.3

Journal of Hospital Infection (2019)²

\$4,817

2.4

Journal of Clinical Microbiology (2019)³

\$4,739

2.0

Journal of Hospital Infection (2011)⁴

\$3,405*

5.4

Journal of Clinical Microbiology (2009)⁵

\$4,142*

1+

Journal of Hospital Medicine (2006)⁶

\$3,647*

3.0

The American Journal of Medicine (1999)⁷

\$6,319**

2.0

Clinical Performance Quality Healthcare (1998)⁸

\$3,962*

8.4

Journal of American Medical Association (1991)⁹

\$3,303*

4.5

The American Journal of Medicine (1989)¹⁰

\$4,431*

4.2

Average Cost Per False-Positive Event

\$4,162

3.4 days

1. Klucher J, Davis K, Lakkad M, Painter JT, Dare RK. Risk factors and clinical outcomes associated with blood culture contamination. Infect Control Hosp Epidemiol. 2022;43(3):291-297. doi:10.1017/ice.2021.111. 2. Geisler BP, Jilg N, Patton RG, Pletzsch JB. Model to evaluate the impact of hospital-based interventions targeting false-positive blood cultures on economic and clinical outcomes. J Hosp Infect. 2019;102(4):438-444. doi:10.1016/j.jhin.2019.03.012. 3. Skoglund E, Dempsey CJ, Chen H, Garey KW. Estimated clinical and economic impact through use of a novel blood collection device to reduce blood culture contamination in the emergency department: a cost-benefit analysis. J Clin Microbiol. 2019;57(1):e01015-18. doi:10.1128/JCM.01015-18. 4. Alahmadi YM, Aldeyab MA, McEnay JC, et al. Clinical and economic impact of contaminated blood cultures within the hospital setting. J Hosp Infect. 2011;77(3):233-6. doi:10.1016/j.jhin.2010.09.033. 5. Gander RM, Byrd L, DeCrescenzo M, Hrary S, Bowen M, Baughman J. Impact of phlebotomy-drawn blood cultures on contamination rates and health care costs in a hospital emergency department. J Clin Microbiol. 2009;47(4):1021-4. doi:10.1128/JCM.02162-08. 6. Zwang O, Albert RK. Analysis of strategies to improve cost effectiveness of blood cultures. J Hosp Med. 2006;1(5):272-6. doi:10.1002/jhm.115. 7. Little JR, Murray PR, Traynor PS, Spitznagel E. A randomized trial of povidone-iodine compared with iodine tincture for venipuncture site disinfection: effects on rates of blood culture contamination. Am J Med. 1999;107(2):119-25. doi:10.1016/S0002-9343(99)00197-7. 8. Surdulescu S, Utamsirgh D, and Shekar S. Phlebotomy teams reduce blood-culture contamination rate and save money. Clin Perform Qual Health Care. 1998;6(2):60-2. 9. Bates DW, Goldman L, Lee TH. Contaminant blood cultures and resource utilization. The true consequences of false-positive results. JAMA. 1991;265(3):365-9. doi:10.1001/jama.1991.03460360365. 10. Dunagan WC, Woodward RS, Medoff G, et al. Antimicrobial misuse in patients with positive blood cultures. Am J Med. 1989;87(3):253-9. doi:10.1016/S0002-9343(89)80146-9. *Adjusted by a 40% cost-to-charger ratio and then inflation adjusted using CPI Inflation Calculator (https://www.bls.gov/data/inflation_calculator.htm) from June year of publication to June 2019. Did not adjust 2022 study given recency. **Inflation adjusted using CPI Inflation Calculator (https://www.bls.gov/data/inflation_calculator.htm) from June 1999 to June 2019.